Semester Project

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July 2026

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1 SMC EX260-SDN1 DeviceNet

1.1 Physical Switch Settings

1.1.1 Communication Speed

The communication speed is set with the physical switches No. 3 and 4. The default value is 125kbps.

Bitrate	Switch No.	
kbps	3	4
125	0	0
250	0	1
500	1	0

Table 1: Bitrates corresponding to Switch Settings

1.1.2 DeviceNet Address

The DeviceNet address is set with the physical switches No. 5 to 10. The default address setting is 63.

Address	Switch No.					
-	5	6	7	8	9	10
0	0	0	0	0	0	0
1	0	0	0	0	0	1
2	0	0	0	0	1	0
...				...		
62	1	1	1	1	1	0
63	1	1	1	1	1	1

Table 2: Addresses corresponding to Switch Settings

1.2 Messaging Protocol

A message consists of a connection ID and a data field. Wait $\geq 1s$ for a response when you send a message.

1.2.1 Connection ID

The connection ID counts 11 bits. The structure depends on the message group. Only group 2 is relevant for this project. The connection ID for group 2 contains the group ID (2 bits), the MAC ID (6 bits) and the message ID (3 bits).

	Group ID		MAC ID						Message ID		
Bit No.	10	9	8	7	6	5	4	3	2	1	0

Table 3: Group 2 Connection ID Structure

Message ID	Function
4	Master's Explicit Request
6	Master/Slave Connection Management
7	Duplicate MAC ID Check

Table 4: Relevant Group 2 Message IDs

1.2.2 Data Field

The data field may include some flags. The R flag indicates a request or response. The F flag indicates a complete or fragmented message. The X flag is a transaction ID echoed in a response to match a request.

Flag	0	1
R	Request	Response
F	Complete	Fragmented
X	-	-

Table 5: Data Field Flags

1.2.3 Setup Flowchart

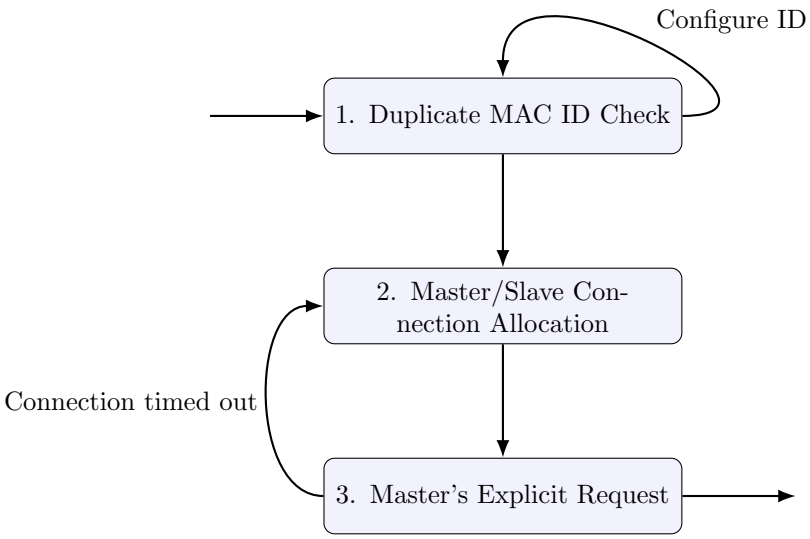


Figure 1: Communication Setup

1.3 Duplicate MAC ID Check

It's inevitable that two modules on the same link will be assigned the same address. It requires human intervention to configure a unique Pi address. The Pi address is written to the Connection ID MAC ID for the duplicate check.

Group ID		Pi Address						Message ID		
1	0	0	0	0	0	0	0	1	1	1

Table 6: Duplicate MAC ID Check Connection ID

Byte No.		
0	R	Physical Port Number
1		Vendor ID
2		
3		Serial Number
4		
5		
6		

Table 7: Duplicate MAC ID Check Data Field

Important: The duplicate MAC ID check has to be transmitted two consecutive times without receiving a subsequent request or response message before establishing a connection.

1.4 Master/Slave Connection Management

Group ID		DeviceNet Address						Message ID		
1	0	1	1	1	1	1	1	1	1	0

Table 8: Master/Slave Connection Management Connection ID

The Pi address is written to the MAC ID in the data field. The service code 0x4B indicates the allocation of a master/slave connection and 0x4C the release. The Class ID 3 directs the message at the DeviceNet object. There exists exactly one DeviceNet object per physical attachement to the DeviceNet. Therefore, the instance ID is always 1. The allocation choice byte denotes an explicit message and I/O connections from the predefined master/slave connection set. The acknowledge suppression bit has to be set to 0 for the release choice.

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4B]	
2		Class ID [0x03]	
3		Instance ID [0x01]	
4		Allocation Choice	
5		Allocator's MAC ID	

Table 9: Master/Slave Connection Allocation Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4C]	
2	Class ID [0x03]		
3	Instance ID [0x01]		
4	Release Choice		

Table 10: Master/Slave Connection Release Data Field

Bit No.	Connection
7	Reserved
6	Acknowledge Suppression
5	Cyclic
4	Change of State
3	Multicast Polling
2	Bit Strobed
1	Polled
0	Explicit Message

Table 11: Allocation/Release Choice

1.5 Master's Explicit Request

Explicit request messages are used to perform operations such as reading and writing attributes. The Pi address is written to the MAC ID in the data field. The service code 0x10 is used to write to a single attribute and the code 0x0E to read from a single attribute. The class ID 9 represents the discrete output points. The instance ID corresponds to the channel.

Group 2		DeviceNet Address						Message 4		
1	0	1	1	1	1	1	1	0	1	0

Table 12: Master's Explicit Request CID

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		
5	Value		

Table 13: Master's Explicit Set Single Value Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		

Table 14: Master's Explicit Get Single Value Data Field

The class ID 4 represents the Assembly Object. The instance ID 37 corresponds to the 32 solenoid output points. The 4 value bytes act as a mask to set the values for all 32 channels at once.

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		
5	Value		
6			
7			
8			

Table 15: Master's Explicit Set Multiple Values Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		

Table 16: Master's Explicit Get Multiple Values Data Field

1.6 Software

1.6.1 Interface

Signature	Description
`duplicate_check()`	Performs the duplicate MAC ID check.
`connect()`	Allocates the master/slave connection.
`disconnect()`	Releases the master/slave connection.
`get_status(id)`	Returns the status bit for a valve.
`get_status_all()`	Returns a list of active valve IDs.
`open(id)`	Opens a single valve.
`open_multiple(idx)`	Opens a list of valves.
`open_all()`	Opens all valves.
`close(id)`	Closes a single valve.
`close_multiple(idx)`	Closes a list of valves.
`close_all()`	Closes all valves.
`start()`	Performs duplicate check and allocates connection.
`stop()`	Closes all valves and disconnects the DeviceNet.

Table 17: Function Signatures and Descriptions

2 Keller PAA-4LD I^{2}C

The Keller addresses are set from 0x40 to 0x49. The absolute pressure ranges from $p_{\min} = 0$ bar to $p_{\max} = 3$ bar.

2.1 Request Measurement

The command to request a measurement is 0xAC. Wait $\geq 8ms$ before reading the measurement.

2.2 Read Measurement

The response contains 3 bytes: The status, the pressure's most and least significant byte. The status carries a checksum error indicator at bit 2. The pressure

p is interpreted with $(p - 16384) * (p_{\max} - p_{\min})/32768 + p_{\min}$.

2.3 Software

2.3.1 Interface

Signature	Description
`get_pressure(id)`	Reads pressure from a sensor.
`get_pressure_all()`	Returns a list of pressures from all sensors.

Table 18: Function Signatures and Descriptions

3 Autonics TM4-N2SB RS485

The default address is 1. The default baudrate is 9600 bps.

3.1 Function Code

Code	Function
1	Read Coil Status
3	Read Holding Registers
4	Read Input Registers
5	Force Single Coil
6	Preset Single Register

Table 19: Relevant Function Codes

3.2 Register Address

Code	Address for Channel i	Function
1/5	0x0002 * (i-1)	Read/Write Status
3/6	0x03E8 * (i-1)	Read/Write Set Temperature
4	0x03E8 + 6 * (i-1)	Read Temperature

Table 20: Functions

Important: Run=0, Stop=1

3.3 Software

3.3.1 Interface

Signature	Description
`get_temperature(id)`	Reads the current temperature for a channel.
`get_temperature_multiple(idx)`	Returns temperatures for a list of channels.
`get_temperature_all()`	Returns a list of temperatures for all channels.
`set_temperature(id, value)`	Sets the target setpoint temperature for a channel.
`set_temperature_multiple(idx, value)`	Sets target temperature for a list of channels.
`get_set_temperature(id)`	Reads the configured setpoint temperature for a channel.
`get_status(id)`	Returns the run/stop status bit for a channel.
`run(id)`	Starts operation for a single channel.
`stop(id)`	Stops operation for a single channel.
`run_multiple(idx)`	Starts operation for a list of channels.
`stop_multiple(idx)`	Stops operation for a list of channels.

Table 21: Function Signatures and Descriptions

4 TCP/IP

5 GUI

The GUI consists of a control panel, the pressure output and a samples table. The other tabs are used to display and edit configurations.

6 Threads

Every device has its own worker thread. All communication to the devices is handled through these workers to avoid interleavings. Procedures, such as conditioning, have their own threads that work in the background to avoid freezing the GUI.